

Normal Harmonic Search Algorithm-Based MPPT for Solar PV System and Integrated With Grid Using Reduced Sensor Approach and PNKLMS Algorithm

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Abstract—This paper deals with a novel reduced sensor strategy, for two-stage single-phase grid connected solar photovoltaic system with a battery, where the dc-link voltage sensor is not required. Moreover, for maximum power point tracking (MPPT), a new “normal harmonic search” (NHS) algorithm is used, and for voltage source converter (VSC) control, a novel “power normalized kernel least mean square” (PNKLMS) algorithm is proposed. The PNKLMS algorithm is used for active load component extraction from the load current, even when the load current is characterized by linear or nonlinear behavior. Moreover, PNKLMS algorithm is used for reference grid current generation. The NHS algorithm is an improved version of the harmony search, which is inspired to the music composition technique and its searching ability is improved by using the normal probability distribution factor. Therefore, NHS reaches the global maximum power point very quickly and is suitable for online searching on hardware. Moreover, the extracted power is fed into the grid. For control of VSC, the proposed PNKLMS control technique is used, which maintains the power quality. The performances of NHS-based MPPT and PNKLMS-based control technique with reduced sensor approach are tested on a developed prototype. The satisfactory steady-state and dynamic performances of the NHS algorithm are demonstrated under variable irradiance and temperature during partial shaded conditions. Moreover, the performances of the PNKLMS algorithm, without using dc-link voltage sensor are demonstrated during overvoltage, undervoltage, harmonics distortion, and load nonlinearity condition, results of which have satisfied the objective of the proposed system and the IEEE-519 standard.

Index Terms—Maximum power point tracking (MPPT), normal harmonic search (NHS), partial shaded condition, power normalized kernel least mean square (PNKLMS), power quality, single phase grid, solar energy conversion.

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I. INTRODUCTION

THE increased environmental issues and demand of low-cost energy have generated enormous interest toward the utilization of renewable energy sources [1] such as wind energy, solar energy, etc. Because of free availability, small in size, and easy to install and use, the solar photovoltaic (PV) energy is popular for residential users as well as industrial users [2], [3]. Here, for reliable operation and optimum uses of the PV panel, maximum power is extracted from the panel as well as quickly fed to the load, and extra power is supplied to the grid [4]. Moreover, a battery is connected on dc link.

According to the property of the battery, the battery voltage is almost constant on every state of charge (SOC), with slight fluctuation, which is handled by voltage source converter (VSC) in a grid connected mode. The VSC converts dc-link voltage very accurately into ac, for synchronization with the grid voltage, through proper adjustment of switching sequence. Therefore, dc-link voltage sensor is not required in a two-stage single-phase grid connected PV system, with a battery on dc link, which provides an extra flexibility to perform well in transient and dynamic conditions. Only two sensors are required for PV voltage and PV current, which is useful in maximum power point tracking (MPPT).

A literature review on the MPPT of PV array reveals that various traditional methods and soft computing techniques have been employed for searching the maximum power point, such as perturb and observe [5], incremental conductance [6], and beta techniques. All these techniques perform well in steady-state condition, but their dynamic performance is not good. In practical situations, solar insolation and temperature are never constant; they are always fluctuating in nature. Therefore, soft computing-based search techniques have been used for searching the maximum power point, such as fuzzy logic [7], neural network, particle swarm optimization [8], intelligent monkey king evolution algorithm [9], flashing fireflies optimization [10], colony of foraging ants algorithm [11], etc. Here, all these techniques can search the global maximum power point (GMPP), but take 1 to 2 s on every dynamic change conditions. However, in practical situation, the solar insolation and temperature vary very rapidly, in the order of less than 1 s. Therefore, there is a need of a strong searching algorithm, which can reach the GMPP very quickly.

Moreover, there is a need for a robust control technique for optimal operation, because the extracted maximum power is fed in nonlinear load and the rest of the power is supplied to the grid. Nonlinear load is connected at the point of common coupling (PCC), which degrades the quality of supply power by injecting harmonics in the supply current. Therefore, load compensation, power factor correction, and power balancing are the main tasks of the control technique.

A literature review on “grid-tied solar energy conversion system” reveals that, recently, various adaptive control techniques have been proposed such as least mean square (LMS) [12], adaptive neuro-fuzzy inference based control [13], delta power control [14], neutral point clamped [15], etc. However, the problems with these techniques are that they are not able to show performance of the system under abnormal conditions, which are the most unusual phenomena in distributed energy conversion system. Few other control techniques are discrete Fourier transform (DFT) [16], second order generalized integrator (SOGI) [17], Prony’s method (PM) [18], frequency locked loop (FLL) [19], Kalman filter (KF) [20], etc. However, DFT requires stationary waveform with fixed length and assumes that the waveform window is repetitive in nature, which is not possible on online searching. SOGI and FLL are highly sensitive in case of dc offset and low-order harmonics. PM creates huge computational burden due to high order of polynomial equation. KF is good for instantaneous estimation by using state variables in two-step (prediction and correction). The KF is further modified for high-frequency system, which is linear KF [21] and extended KF [22]. However, the problem with these techniques is the heavy computational burden during the linearization process because, in linearization, derivative terms are used, which is the main source of computational complexity and burden. Some research works have revealed that some modified LMS techniques have been proposed where computational complexity and burden are relatively low, such as combined LMS, least mean fourth-based control, and recursive least square-based control [23]. However, in these techniques, the convergence speed is compromised, which is enhanced in normalized LMS [24] technique. In [24], the considered input signals are already in a normalized state, so proposed enhancement is not granted.

Therefore, in this paper, a novel power normalized kernel LMS (PNKLMS) algorithm-based control technique is proposed for the robust control of grid connected energy conversion system. Here, the computational burden is low, because of free from derivative term, as well as the performance of PNKLMS algorithm is instantaneous and suitable for high-frequency system, because of hybridization of power normalized kernel function with kernel LMS algorithm.

Moreover, for MPPT, a novel normal harmonic search (NHS) algorithm is proposed for all types of weather conditions. This algorithm is an improved form of the harmony search (HS) [25], which is inspired to the music composition technique and its searching ability is improved by using the normal probability distribution function [26]. The highly appreciable merits of NHS algorithm [27] are the requirement of small population and performance does not depend on the initial value as well as a dependency on the algorithm specified parameters is very less.

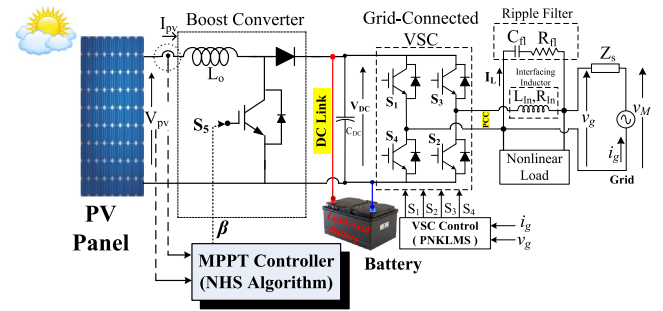


Fig. 1. Two-stage single-phase grid-connected battery supported solar PV system.

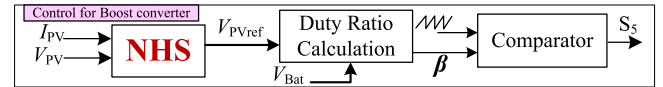


Fig. 2. Control scheme for dc-dc converter.

Therefore, it is highly suitable for online searching and creates a very less computational burden. In this work, these merits of NHS algorithm are demonstrated on the fluctuating solar insolation and temperature conditions. Moreover, performances of PNKLMS algorithm are demonstrated under overvoltage, undervoltage, harmonic distortion, and load nonlinearity condition.

II. SYSTEM CONFIGURATION

The proposed “battery supported two-stage single-phase grid connected solar PV system with load” is shown in Fig. 1.

In this system, two step power conversion processes is used. In first step, a boost converter is used for maximum power extraction from PV array, and extracted power is fed on dc link. For controlling of the boost converter, NHS algorithm is used, which generates optimum duty cycle (β) for switching of boost converter. On dc link, a lead-acid battery is attached, which automatically charges when excess power is available and discharges when load requires, as well as it maintains the dc-link voltage. In second stage, a VSC is used, which converts dc-link power into ac form, and fed to the load and rest of the power is supplied to the grid after load is fed. This complete process is controlled by PNKLMS-based control technique, major task of which is load compensation, power quality improvement of supply power, and maintaining unity power factor. The load is attached on PCC, which is in-between VSC and the grid. VSC and the grid are interfaced through an interfacing inductor (L_{ln} , R_{ln}). Moreover, ripple filter (C_{fl} , R_{fl}) is attached on PCC for absorbing switching ripples.

III. CONTROL APPROACH

The control schemes for two-stage single-phase grid-connected solar PV system are shown in Figs. 2 and 3.

In these schemes, NHS algorithm is used for MPPT and PNKLMS algorithm is used for reference current generation as well as load component extraction.

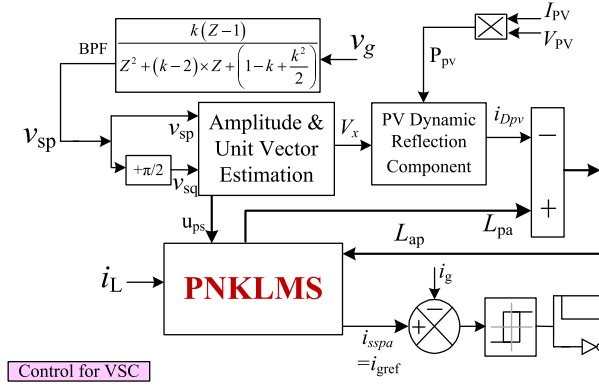


Fig. 3. Control scheme for VSC.

First, NHS algorithm is used for MPPT, which generates a reference PV voltage (V_{PVref}). According to V_{PVref} , an optimal duty cycle (β) for the maximum peak is generated, by keeping constant battery voltage (V_{Bat}). β is calculated as

$$\beta = 1 - \frac{V_{PVref}}{V_{Bat}}. \quad (1)$$

β is compared with the show-tooth wave for generating signal (S_5) for the boost converter. After this, PNKLMS-based control technique is used for VSC. Here, grid voltage (v_s) is filtered by using a bandpass filter (BPF), which output is in-phase quantity (v_{sp}) of grid voltage. The open-loop transfer function of the BPF is given as

$$T_f = \frac{k(Z-1)}{Z^2 + (k-2)Z + (1-k + \frac{k^2}{2})} \quad (2)$$

where $k = \sqrt{2} \times \omega \times T_s$, ω is natural frequency ($2\pi f$), and T_s is sampling time period.

The quadrature component (v_{sq}) of v_s is generated by shifting $+\pi/2$. The amplitude of grid voltage (V_x), and in-phase unit-template (u_{ps}) are calculated as

$$V_x = \sqrt{v_{sp}^2 + v_{sq}^2} \quad (3)$$

$$u_{ps} = \frac{v_{sp}}{\sqrt{v_{sp}^2 + v_{sq}^2}}. \quad (4)$$

For dynamic performance improvement, the reflection of the instantaneous change in PV power (P_{PV}) on grid current (i_g) is taken under consideration by using PV dynamic reflection component (i_{Dpv}). i_{Dpv} is derived as

$$i_{Dpv} = \frac{2 \times P_{PV}}{\sqrt{v_{sp}^2 + v_{sq}^2}}. \quad (5)$$

The ac loss component (β_{AC}) is calculated from V_x and reference of terminal voltage (V_{xn}). The difference of V_x and V_{xn} generates ac-link error ($e_{AC} = V_{xn} - V_x$). e_{AC} is fed to a PI controller, which generates ac loss component (β_{AC}). β_{AC} is described as

$$\beta_{AC}(n+1) = G_P e_{AC}(n) + G_I \int_0^t e_{AC}(n) dn \quad (6)$$

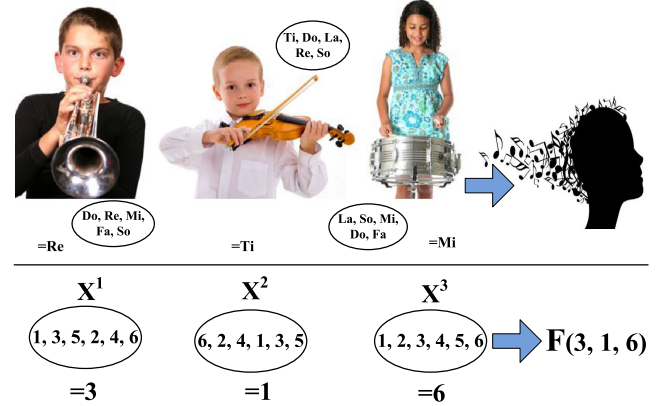


Fig. 4. Pictorial representation of HS algorithm.

where G_I and G_P are integral and proportional gains of PI controller, which regulates ac voltage.

For generating active (i_{ssa}) component of reference grid current, the PNKLMS control is used. For i_{ssa} generation, load current (i_L) and u_{ps} are used, which also generate estimated active weight component (L_{pa}). The generation process of L_{pa} and i_{ssa} is represented as

$$[L_{pa}, i_{ssa}] = f_{PNKLMS}(i_L, u_{ps}). \quad (7)$$

The resultant estimated active weight component (L_{ap}) is calculated as

$$L_{ap} = L_{pa} - I_{Dpv}. \quad (8)$$

The reference grid current (i_{gref}) is generated as

$$i_{gref} = i_{ssa}. \quad (9)$$

By using hysteresis controller (HC), the switching pulses (S_1, S_2, S_3 , and S_4) are generated, where the input signals are grid current (i_g) and i_{gref} . The HC is a current controller, which shapes the current of the inverter. The advantages of using HC are its hysteresis band, which gives an excellent dynamic performance, and ability to control the peak to peak value of current ripple in desired hysteresis band limit.

A. NHS Algorithm

The NHS algorithm is the improved version of the HS by using normal probability distribution factor. This improves the searching ability and enhances the rate of convergence [27].

1) *Harmony Search (HS)*: HS is inspired from the music composition technique. The main ideology of this searching technique is searching or selecting a perfect harmony or rhythm for a particular song. In MPPT, the main aim is find a perfect duty cycle (harmony) for a particular GMPP (song) [25]. The pictorial representation of HS is shown in Fig. 4.

The major stages during execution are harmony memory (HM) allocation, adjustment of pitch and randomization.

- 1) Harmony memories are the set of duty cycles according to the power of the individuals. Here, the duty cycles or harmony memories are allocated according to HM acceptance rate (ψ_{HMAR}). This ψ_{HMAR} is $\in [0,1]$.

- 2) Pitch adjustment is dependent on two factors: bandwidth (ω_k) and pitch adjustment rate (λ_{PAR}^k) [25], both are defined as

$$\lambda_{PAR}^k = \lambda_{\min} + \frac{\lambda_{\max} - \lambda_{\min}}{K} \times k \quad (10)$$

$$\omega^k = \omega_{\max} \times e^{\frac{\ln(\frac{\omega_{\min}}{\omega_{\max}})}{K} \times k} \quad (11)$$

$$P^{k+1} = P^k + \omega^k \bullet \varphi \quad (12)$$

where P^k and P^{k+1} are previews and new pitch, respectively, and φ is the random number (range of $[-1, +1]$). $\lambda_{\min}$ and $\lambda_{\max}$ are the minimum and maximum pitch adjusting rate, respectively. $\omega_{\min}$ and $\omega_{\max}$ are the minimum and maximum bandwidth, respectively. K is the total number of iteration and k is the iteration number.

- 1) Randomization: It enhances the diversity of the solution. Moreover, by using the probability of randomization (η_{ran}) and pitch coefficient (η_{pitch}), it drives the system behavior, to further explore global optimality. These are described as

$$\eta_{\text{ran}} = 1 - \psi_{\text{HMAR}} \quad (13)$$

$$\eta_{\text{pitch}} = \psi_{\text{HMAR}} \times \lambda_{PAR}^k. \quad (14)$$

2) *Normal Harmony Search*: In NHS, the normal probability distribution factor (Gaussian distribution factor) (ε') multiplies with bandwidth of HS to further enhance the convergence efficiency [27]. This factor reduces the requirement of large population and the number of iterations. This factor [26] is based on two parameters (variance (σ) and expectation of the distribution (τ)) as

$$\varepsilon' = \frac{1}{\sigma \times \sqrt{2\pi}} e^{-\frac{(x-\tau)^2}{2 \times \sigma^2}}. \quad (15)$$

The solution (duty cycle (d_j^k)) updating process for j th element in k th iteration is formulated as

$$\begin{aligned} d_j^{k+1} &= d_j^k \pm \varepsilon' \times \omega^k & \text{if } \varphi_{\text{new}} < \lambda_{PAR}^k \\ d_j^{k+1} &= d_j^k & \text{if } \varphi_{\text{new}} \geq \lambda_{PAR}^k. \end{aligned} \quad (16)$$

The pseudo code of the NHS algorithm for MPPT problem is given as follows.

Normal Harmony Search
begin (NHS)
Generate initial duty cycles (harmonics)
Receive corresponding voltage and current
Objective fitness function $P_k(t)$
Define λ_{PAR}^k , Pitch limits, ω , σ and λ
Define ψ_{HMAR}
while ($k < K$)
Generate new harmonics (duty cycle) by accepting best harmonics
Adjust pitch to get new duty cycle (harmonics)
if ($\varphi > \psi_{\text{HMAR}}$), select an existing duty cycle (harmonics) randomly

Continued

else if ($\varphi > \lambda_{PAR}^k$), adjust the pitch factor randomly within $[0, 1]$

else generate new duty cycle (harmonics) through randomization

 update by using Normal factor

end if

end if

 Accept the new duty cycle (harmonics) if good

end while

 decide and store the current best duty cycle

 Check for stopping criteria ($\Delta P < 0.001$)

if (NO), go to NHS

else find the GMPP

end

end (NHS)

B. Power Normalized Kernel Least Mean Square

In PNKLMS algorithm, power normalized kernel trick (PNKT) [26] is hybrid with LMSs algorithm. PNKT is used to map the input signal into a high-dimensional space (HDS), which finds linear relations in the input signals. It helps the control algorithm to handle the input signal in the HDS, without knowing the coordinates of the signal in that space. Here, instead of calculating the inner products, the kernel of input signal can map the input signal into the HDS simply by computing. Here, by using PNKLMS, the prediction error (ϕ_{ape}) of active component is calculated as

$$\phi_{\text{ape}}(n) = i_L(n) - u_{ps}(n) \times L_{pa}(n). \quad (17)$$

ϕ_{ape} is minimized through proper updating of $L_{pa}(n)$. Updating process of $L_{pa}(n)$ is described as

$$L_{pa}(n+1) = L_{pa}(n) + \frac{2\mu_p(n)}{\langle \mu_p(n), \sigma_p(n) \rangle} u_{ps}(n) \times \phi_{\text{ape}}(n) \quad (18)$$

where $\langle \mu_p(n), \sigma_p(n) \rangle$ is power kernel function, which is calculated as

$$\langle \mu_p(n), \sigma_p(n) \rangle = K_f(\mu_p(n), \sigma_p(n)) = 1 - \frac{\|\mu_p(n) - \sigma_p(n)\|}{\varepsilon} \quad (19)$$

By using the scaling factor, the power kernel function is normalized as

$$K_{fn}(\mu_p(n), \sigma_p(n)) = 1 - \delta \times \frac{\|\mu_p(n) - \sigma_p(n)\|}{\varepsilon} \quad (20)$$

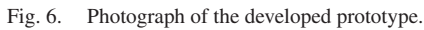
where ε and δ are scaling factor ($7.5 < \varepsilon < 10$) and normalizing factor ($0.001 < \delta < 0.01$). $\sigma_p(n)$ is autocorrelation factor of the active component, which is described as

$$\sigma_p(n+1) = \vartheta \times \sigma_p(n) + (1 - \vartheta) \times \phi_{\text{ape}}(n) \times \phi_{\text{ape}}(n-1). \quad (21)$$

$\mu_p(n)$ is active weight constant, which is calculated as

$$\mu_p(n+1) = \tau \times \mu_p(n) + \varsigma \times (\sigma_p(n+1))^2 \quad (22)$$

where, τ ($\tau < 0.01$) and ς ($\varsigma < 0.0001$) are accelerating parameters.


$$I_{sspa} = L_{ap} \times u_{ps} \quad (23)$$

IV. RESULTS AND DISCUSSION

During testing, an actual grid is used with a PV simulator (ETS600x17DPVF, AMETEK), and for operating of PV array, on maximum power point, a boost converter is used as well as for feeding load and synchronization with the grid, a VSC is used. For reducing the effect of switching ripples and harmonics, interfacing inductor and RC filter are used. For the execution of control algorithm, a digital signal processing controller (1103-DSPACE) is used, input of which are signals from voltage (P-LV25) and current sensors (P-LA55). The system parameters are given in Table I.

In this section, it is considered that the PV generated power is more than the required load power, so the extra power is fed into

Parameter	Value	Parameter	Value	Parameter	Value
I_{sc}	15A	P_{Load}	1.16kW	v_g	220v
V_{oc}	350V	L_{in}	2.5mH	f	50Hz
V_{mpp}	301.42V	C_{fl}	12 μ F,	δ	0.01
I_{mpp}	14.2A	R_{fl}	5 Ω	τ	0.001
P_{mpp}	4.286kW	ϵ	10	ζ	0.00001

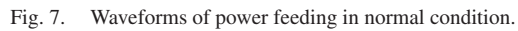


Fig. 7 depicts that the power is fed into the grid, because the waveforms of v_g and i_g are out of phase, also this is shown in Fig. 9(a). The waveform of load voltage and current is shown in Fig. 9(e). The amount of supply power into the grid and load, are shown in Fig. 9(b) and (f), respectively, as well as harmonic spectra of grid voltage, grid current, and load current are shown in Fig. 9(c), (d), and (g), which reveals that 1.88-kW power is fed into the grid and 1.16-kW power is fed into the load, where during it THDs of v_q and i_q are only 3.6% and 2.6%, while

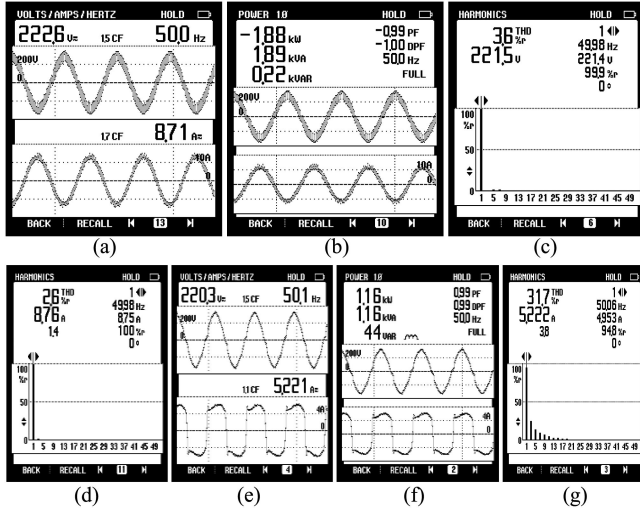


Fig. 9. Waveforms of (a) grid voltage and supply current, (b) power fed into the grid, (c)–(d) harmonic spectra of grid voltage and current, (e) load voltage and current, (f) power fed into the load, and (g) harmonic spectrum of load current.

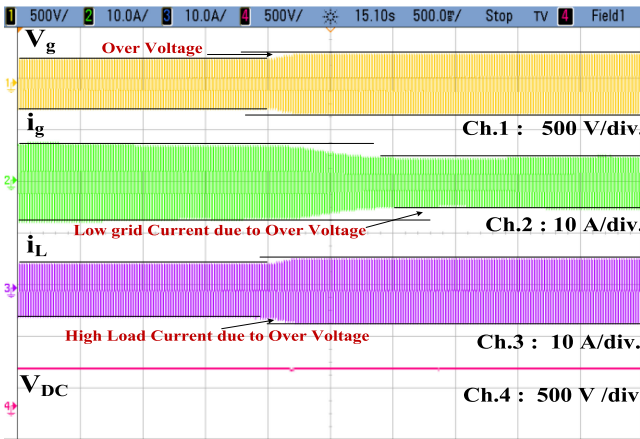


Fig. 10. Waveforms during overvoltage condition.

THD of i_L is 31.7%. Moreover, Fig. 7 shows the balance dc-link voltage and nonlinearity of the load, which can be seen in the waveform of load current.

In Fig. 8, the waveforms of PV voltage and current at 1000 W/m² solar irradiance are shown. Moreover, by using PN-KLMS control technique, the extracted fundamental component ($i_{sspa}(n)$) and active load component ($L_{Pa}(n)$) from nonlinear load current are shown in Fig. 8, which shows, good filtering ability of PNKLMS algorithm.

2) *Operation During Grid Overvoltage Condition:* During overvoltage, the sudden rise of 11% in grid voltage is considered, waveforms of which are shown in Figs. 10 and 11.

Figs. 10 and 11 depict that during overvoltage, the voltage at PCC is increased, so the grid current is decreased, and load current is increased, because load power is proportional to the square of PCC voltage ($P_L \propto V_{PCC}^2/Z_L$). Moreover, the THDs of i_g and v_g are still low, as well as balanced and constant power is fed into the load and an extra power is supplied to the grid.

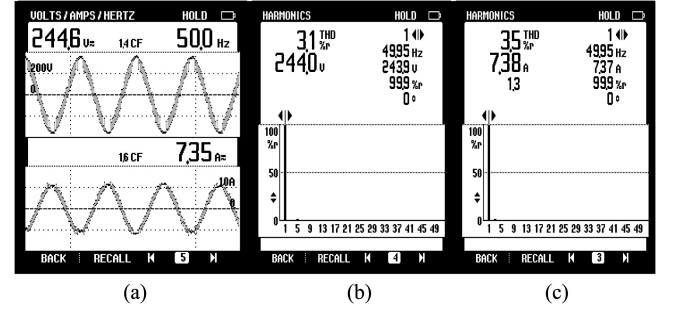


Fig. 11. Waveforms of (a) grid voltage and current, (b)–(c) harmonic spectrum of grid current, and grid voltage, during grid overvoltage condition.

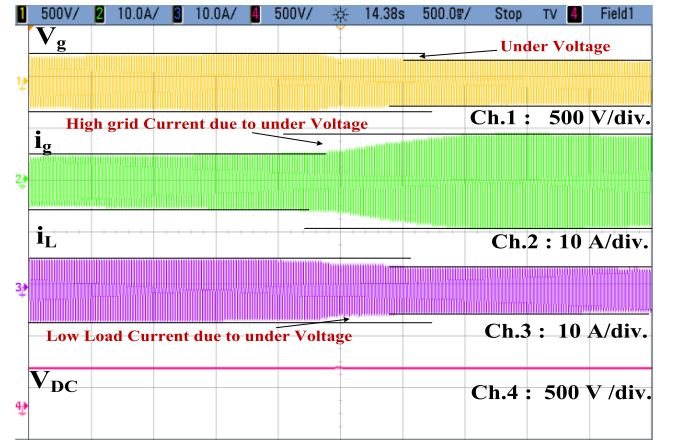


Fig. 12. Waveforms during under-voltage condition.

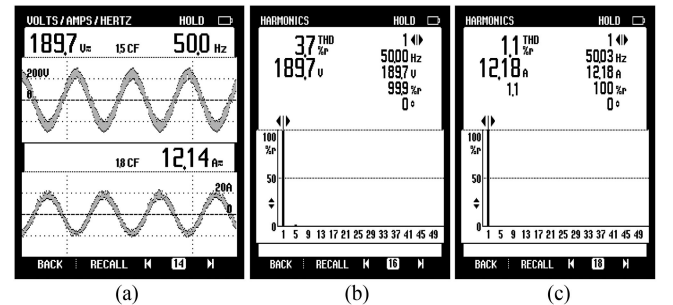


Fig. 13. Waveform of (a) grid voltage and current, (b)–(c) harmonic spectrum of grid current and grid voltage, during grid under voltage condition.

3) *Operation During Grid Undervoltage Condition:* During undervoltage, sudden fall of 15% in the grid voltage is considered, waveforms of which are shown in Figs. 12 and 13.

Figs. 12 and 13 depict the performance during under voltage. The voltage at PCC is decreased, so the grid current is increased, and load current is decreased, because the load power is proportional to the square of PCC voltage. Moreover, the THDs of i_g and v_g are still low, as well as a constant power is fed into the load and extra power is supplied to the grid.

4) *Operation Under Distorted Grid Voltage Condition:* During distorted grid voltage, a 21.9% THD in grid voltage is con-

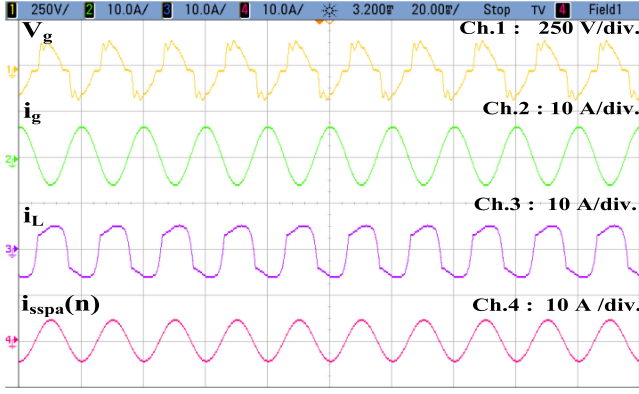


Fig. 14. Waveforms under distorted grid voltage condition.

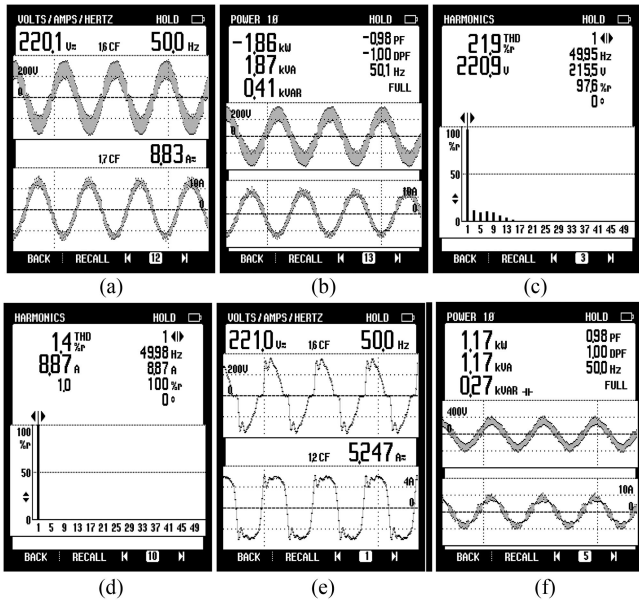


Fig. 15. Waveforms of (a) grid voltage and current, (b) power supply into the grid, (c)–(d) harmonic spectrum of grid voltage and current, (e) load voltage and current, and (f) power supply into the load during distorted grid voltage condition.

sidered, and in this situation, power is fed successfully into the grid, waveforms of which are shown in Figs. 14 and 15.

Figs. 14 and 15 show that in distorted grid voltage condition, PNKLS control algorithm has extracted fundamental component very accurately and it feeds sinusoidal grid current. Moreover, the power is still constant, which can be seen in Fig. 15(a) and (b). Therefore, the THDs of v_g and i_g are 21.9% and 1.4%, which can be seen in Fig. 15(c) and (d), respectively. The waveforms of nonlinear load current, distorted load voltage, and supply load power are shown in Fig. 15(e) and (f).

B. Operation Under Solar Irradiation Change Condition

During irradiation change condition, the performance of the proposed NHS algorithm is closely observed and evaluated. The results are compared with the results of HS algorithm, as well as true maximum power point.

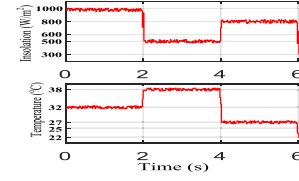


Fig. 16. Pattern of insolation and temperature variation.

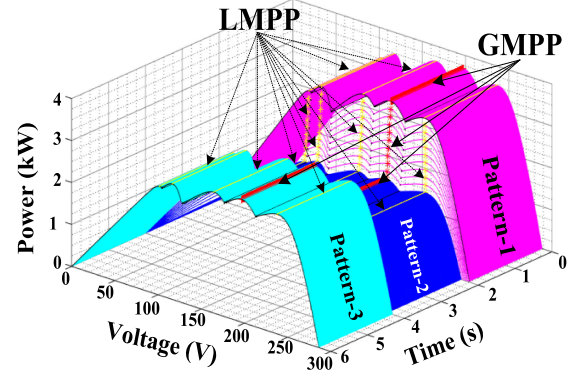


Fig. 17. Pattern of PV curves.

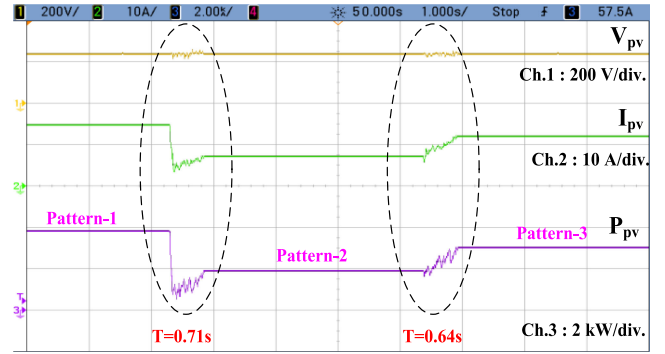


Fig. 18. Experimental results achieved by NHS.

The parameters of the NHS are $\lambda_{\min} = 0.001$, $\lambda_{\max} = 0.999$, $\omega_{\min} = 0.001$,

$$\omega_{\max,k} = 1 / (20 \times (P_{k,\max} - P_{k,\min}))$$

$$\psi_{\text{HMAR}} = 0.80 \text{ and } \lambda_{\text{PAR}} = 0.5.$$

The performances of NHS algorithm and HS algorithm are tested at the highly fluctuating atmospheric conditions. The duration for dynamics change, is taken as 2 s. Here, the variations in temperature and the solar insolation are considered. The patterns of insolation and temperature variation are shown in Fig. 16. Moreover, the patterns of P-V curve are shown in Fig. 17.

The PV array output voltage, current and power waveforms achieved by NHS MPPT algorithm and HS MPPT algorithm are shown in Figs. 18 and 19, respectively.

In experimental results, the time division on the X-axis is 1 s/Div. Figs. 18 and 19 clearly show that during dynamic change condition from pattern-1 to pattern-2, for maximum power point tracking, HS has taken 1.60 s, where NHS algorithm has taken only 0.71 s. Similarly, during dynamic change condition from

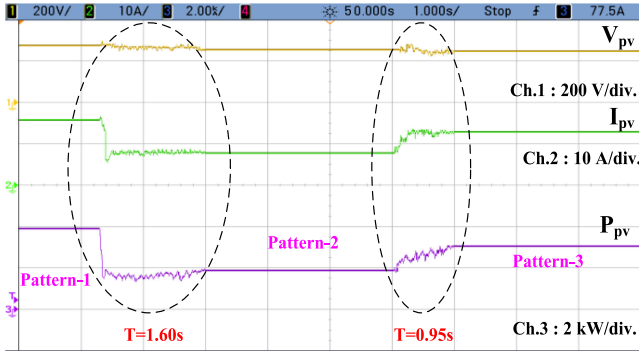


Fig. 19. Experimental results achieved by HS.

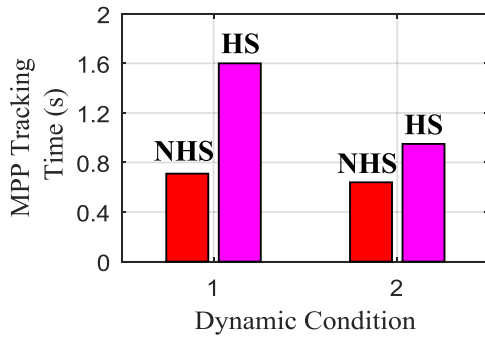


Fig. 20. Bar chart of the tracking time.

pattern-2 to pattern-3, for maximum power point tracking HS has taken 0.95 s, where NHS algorithm has taken only 0.64 s. From these performances, it is cleared that the MPPT performance of NHS, is very good w.r.t. HS. For the comparative analysis of NHS and HS algorithm, the bar chart of the tracking time is shown in Fig. 20.

Moreover, on every insolation change, the NHS algorithm-based global MPPT converges very quickly and takes on an average 0.67 s, as well as, the achieved power at maximum power point is approximately same to the actual maximum power point. Here, the error is 0.013%. This performance reveals that the NHS-based global MPPT searching algorithm is a very good technique to perform in every type of environmental conditions.

C. Operation With Solar PV Generation to Without Solar PV Generation Condition

The solar PV generation to without solar PV generation condition: During solar PV power availability condition, PV power is fed into the load, and after feeding the load, the rest of the power is fed into the grid. Moreover, during without solar PV generation condition, PV power is zero, then it behaves like Distribution Static Compensator (DSTATCOM) and total required load power is supplied by the grid. All these performances are shown in Figs. 21, 22, and 23.

Figs. 21 and 22 show the two modes of operation on a single window. During solar PV generation condition, the PV power is fed to the load and after feeding load, the extra power is fed to the grid. Here, the power is fed into the grid, so v_g and i_g are out of phase from each other, which is shown in

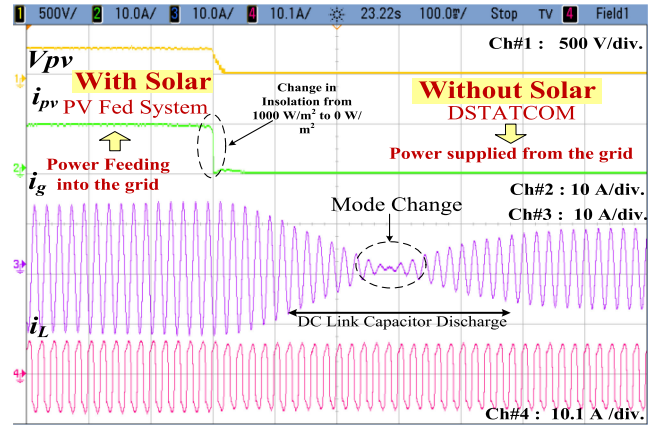


Fig. 21. Waveforms for with solar-to-without solar condition.

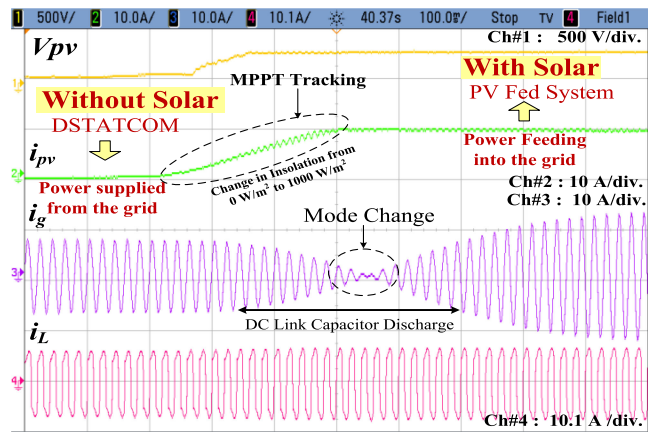


Fig. 22. Waveforms for without solar-to-with solar condition.

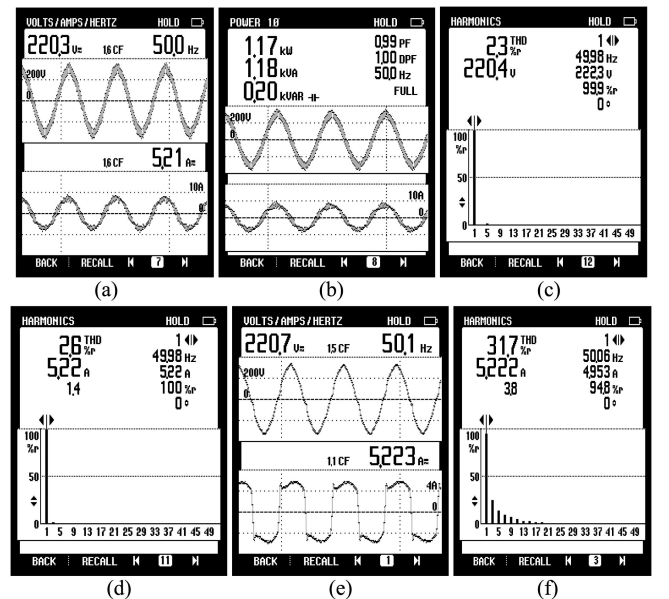


Fig. 23. Waveforms during DSTATCOM operation. (a) Grid voltage and current. (b) Supply power. (c) Harmonic spectrum of grid voltage. (d) Harmonic spectrum of grid current. (e) Load voltage and current. (f) Harmonic spectrum of load current.

Figs. 21 and 22. However, in case without solar PV generation condition, PV power is zero, then it works as DSTATCOM, total required load power is supplied by the grid. Therefore, v_g and i_g are in phase from each other, which is also shown in Figs. 21 and 22. In Fig. 21, the dynamic performance of mode change operation for with to without solar condition is shown. Similarly, in Fig. 22, the dynamic performance of mode change operation for without to with solar PV generation condition is shown. In transient duration, during mode change, dc-link capacitor momentarily supports to maintain the load power, so the load current is constant in every situation, which is shown in Figs. 21 and 22. Moreover, in both conditions, THD is very low, which is shown in Fig. 23. Therefore, the overall performance of PNKLMS control algorithm meets the objectives of the work.

V. CONCLUSION

A new reduced sensor scheme for two-stage single-phase grid connected battery supported solar PV system has been proposed, where dc-link voltage sensor is not required. Moreover, for MPPT, an NHS algorithm has been proposed, which is inspired to the music composition technique and its searching ability is improved by using the normal probability distribution factor, as well as for controlling purpose, PNKLMS-based control technique has been proposed. Here, the battery is attached on dc-link, which provides dc-link voltage stability and improves performance during transient and dynamic conditions. Here, the NHS-based GMPP tracking and PNKLMS-based control technique have been tested on reduced sensor two-stage single-phase grid connected partially shaded solar PV energy conversion system on a developed prototype. The satisfactory steady-state and dynamic performances of the NHS algorithm are demonstrated under variable irradiance and temperature during partial shaded conditions as well as performances of the PNKLMS are demonstrated under overvoltage, undervoltage, harmonic distortion, and load nonlinearity condition. Moreover, on circuitry and economics levels, this system without dc-link sensor is reliable, cheap, and easy in maintenance, as well as due to the PNKLMS-based control technique, the computational burden is also very less, so it is easy to implement on the low-cost microcontroller.

APPENDIX

A. Design of Ripple Filter

An LC , second-order low-pass filter is used here for ripple elimination on PCC, which eliminates all higher order switching harmonics. For meeting the IEEE-519 standard, the cutoff frequency (f_c) is kept 1/10th of the inverter switching frequency (f_{sw}). The maximum voltage drop across the inductor less than 3% of the inverter output voltage (v_{vsc}) is considered [29]. The value of all components of LC filter is calculated as

$$f_c = \frac{f_{sw}}{10}$$

$$L_{In} = \frac{0.03 \times v_{vsc}}{2\pi \times f \times I_{vsc\max}} = 2.042 \text{ mH}$$

$$C_{fl} = \frac{1}{(2\pi f_c)^2 L_{In}} = 12.679 \mu\text{F}$$

where L_{In} and C_{fl} are inductor and capacitor of the LC filter. $I_{vsc\max}$ is the maximum rms value of the VSC current. f is the frequency of the output current.

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